

SHUBHAM KUMAR YADAV

AI-NATIVE FULL-STACK ENGINEER · LLM, RAG & AI AGENTS · CLOUD & DEVOPS

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coshubham.github.io/Portfolio- | github.com/coshubham | [LinkedIn](#)

Core skills: Python | Machine Learning | LLM and Vector Search | FastAPI | REST API | React | TypeScript | PostgreSQL | OpenSearch | Docker | Azure | AWS | CI/CD

SUMMARY

AI-native full-stack engineer shipping production AI end to end: LLMs, RAG, AI agents, vector search, APIs and cloud. MCA in Artificial Intelligence (CGPA 8.71) with 11+ projects delivered to production for ATH Infosystems and its clients. Technical owner of bCloud AI Search (bcloud.ai), a multi-tenant AI product-search SaaS on Azure: 70+ FastAPI endpoints, OpenAI embeddings with OpenSearch retrieval, voice and photo search, LLM agents and two React dashboards. Restored a client's dead AWS marketplace search platform (255,350 products, zero errors, frozen backend) and shipped a nightly scikit-learn risk model at 0.936 ROC AUC. Lead a small development team and work directly with clients. Open to relocation and remote roles.

EXPERIENCE

ATH Infosystems, Noida — Full Stack Engineer (AI/ML, DevOps & Cloud)

Dec 2025 – Present

- Delivered 11+ projects to production for ATH Infosystems and its clients — AI search, e-commerce integrations, analytics and BI, FinTech, identity automation, lead capture and websites — owning the technical side from planning to release.
- Technical owner of bCloud AI Search (bcloud.ai), ATH's commercial multi-tenant AI product-search SaaS: planned the architecture and technical roadmap, built and deployed it, and run it in production — 70+ REST API endpoints across ingestion, search, analytics, billing, admin and client-portal services in Python and FastAPI, two React dashboards, and Docker Compose running OpenSearch, PostgreSQL and Redis behind nginx and Let's Encrypt TLS on an Azure VM I provisioned.
- Engineered the semantic search core with OpenAI text-embedding-3-large (3072-dimension vectors) indexed into OpenSearch, so plain-language queries such as "gift for someone who cooks" return relevant cookware instead of keyword matches; cut search latency from ~500 ms to ~150-250 ms (repeat queries ~0 ms) with a two-tier embedding cache, single-pass query execution and off-event-loop search calls, and shrank responses from 83 KB to 19 KB with gzip.
- Built an in-product AI helpdesk agent on an OpenAI tool-calling loop (add synonym, remove synonym, raise support ticket) and enforced multi-tenancy at the API layer — every call scoped by client_id, verified by a 31-test tenant-isolation suite.
- Restored a large e-commerce marketplace client's dead AWS-based search platform onto Azure under a strictly frozen backend, re-indexing 255,350 products with zero indexing errors in 1 hour 52 minutes.
- Built a nightly machine learning withdrawal-risk model for an education-sector client in Python, PySpark and scikit-learn (logistic regression) reaching 0.936 ROC AUC and scoring 1,300+ students into a Delta Lakehouse, and authored 55+ DAX measures wired into a natural-language AI assistant so business users get answers computed live from governed measures.
- Took over sole developer ownership of SplitIT, a FinTech client's live production platform, after the original developer left; own its codebase, AWS infrastructure, backups and technical client responses, and ran the client-approved production pause.
- Lead a small development team — plan and split the technical work, review code and mentor the team — and run client communication from requirements to support; wrote a 51-page technical handbook, integration guides and runbooks.

Salesqueen Software Solutions, Chennai — Front-End Developer (Internship)

Mar 2025 – Jul 2025

Remote, alongside final-semester MCA

- Built responsive, accessible UI components across six delivered products, including a login portal, production portal, Kanban board, e-commerce website and an online cab-booking app.
- Deployed and maintained releases on live servers and kept the interface layer consistent across the product suite.

CSRBOX Foundation via IBM SkillsBuild — AI Intern (Remote)

Jul 2024 – Aug 2024

- Built a conversational chatbot using IBM Watsonx AI and NLP techniques.

Gautam Buddha University, Greater Noida — Admission Department Intern

Jun 2024 – Aug 2024

- Managed a 3,000+ student record dataset in MySQL plus CRM data through the admission cycle, and built an online Bonafide Certificate system that issued certificates from those records.

EDUCATION

Gautam Buddha University — MCA (Artificial Intelligence), CGPA 8.71

Passed

Veer Narmad South Gujarat University — BCA, CGPA 7.61

Passed

CERTIFICATIONS

- DevOps and Cloud Computing — PW Skills · [Certificate](#) · AI Internship, Watsonx AI — IBM SkillsBuild / CSRBOX · [Certificate](#)
- Front-End Developer — Salesqueen · [Certificate](#) · Internship — GBU Admission Department · [Certificate](#)

LANGUAGES

- English, Hindi

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL, DAX, Bash, PHP, Kotlin, HTML5, CSS3
- **AI / Machine Learning & Search:** machine learning, OpenAI embeddings, Large Language Models (LLM), LLM tool-calling and AI agents, prompt & tool design, RAG (Retrieval-Augmented Generation), semantic and vector search (k-NN), OpenSearch (Elasticsearch-compatible), search relevance tuning, vision models and image search (perceptual hashing), speech-to-text and text-to-speech, scikit-learn, PySpark, IBM Watsonx AI, NLP
- **Backend & APIs:** FastAPI, Flask, Node.js, Express.js, Python asyncio, REST API design, OpenAPI / Swagger, GraphQL, JWT, OAuth2, webhooks, multi-tenant API architecture, Stripe API, PayPal API, BigCommerce API, Microsoft Graph API, SCIM
- **Frontend & Mobile:** React, Vite, Tailwind CSS, Bootstrap 5, Recharts, Chart.js, Three.js / WebGL, Web Speech API, responsive and accessible UI, Android (Kotlin, Firebase)
- **Cloud & DevOps:** Azure (VMs, Logic Apps, Key Vault, Entra ID, DevOps), AWS (EC2, ECS Fargate, Lambda, SQS, RDS, S3, CloudFront, IAM, OpenSearch Serverless, ElastiCache, Secrets Manager), Terraform, Docker, Docker Compose, Ansible, GitHub Actions, CI/CD, nginx, TLS/certbot, Linux (Ubuntu, cron)
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server, MongoDB (Atlas), Redis, SQLite, Firebase
- **BI & Data:** Microsoft Fabric, Power BI, DAX, Tabular Editor, Delta Lake / OneLake, Copilot Studio
- **Tools & Practices:** Git, GitHub, Postman, Figma, technical writing & runbooks, team leadership, code review, mentoring

KEY PROJECTS

bCloud AI Search — Multi-Tenant AI Product-Search SaaS

ATH Infosystems

[bcloud.ai](#) · ATH's own product · live in production · technical owner: architecture, planning, build and operations

Planned, built and now run ATH's multi-tenant AI search for online stores: OpenAI text-embedding-3-large (3072-dim) vectors with OpenSearch k-NN behind 70+ FastAPI endpoints, and five ways to search — typed; voice with a spoken LLM answer; conversational queries through an LLM intent step that keeps price and stock limits; multi-product queries such as “laptop and headphones”; and photo search with a vision model plus image fingerprints. Also BigCommerce catalogue sync, a 5-minute trending-score job, an AI helpdesk agent, and 122 automated tests.

bCloud Admin Console & Client Portal

ATH Infosystems

React · Vite · Tailwind CSS · Recharts · the two dashboards of bCloud AI Search

A 12-page admin console (clients, field mapping, OpenSearch, Azure, AI, monitoring, billing, support tickets, search analytics) and an 11-page light/dark client portal where each store tracks its search analytics, sync runs, billing with a month-end forecast, and “missed searches” with one-click synonym fixes checked live against its own catalogue.

Legacy Marketplace Search Platform — Restore & Modernisation

Large e-commerce marketplace client · ATH Infosystems

Engineered a local TLS/SigV4 compatibility shim so code written for AWS OpenSearch Serverless ran unmodified against self-hosted OpenSearch (zero changes to the client's code). Built a FastAPI service powering two new React dashboards behind nginx, wrote idempotent bash installers that redeploy the stack in one command, and fixed a silent analytics event-loss bug.

Predictive Analytics & Reporting Platform — Power BI and Microsoft Fabric

Education-sector client · ATH Infosystems

Enrolment forecasting, retention and admissions analytics: layered semantic models, cross-source CRM-to-SIS integration with row-level security, and an AI assistant answering plain-English questions from governed DAX measures.

AI Lead-Capture Assistant

ATH Infosystems

Built a Flask chat service and embeddable JavaScript widget live on the company's two public sites: answers visitor questions through an LLM and captures qualified leads straight into SQL Server, with a lead admin console, CSV export and Swagger-documented APIs. Authored 44 of the project's 60 commits.

SplitIT — FinTech Income-Splitting SaaS

FinTech client · live production · inherited ownership · ATH Infosystems

Took over a live platform that splits a user's incoming income across payees by rules and pays them out through Stripe Connect and ACH debits, with PayPal Billing subscriptions — React, TypeScript, Express and PostgreSQL on AWS (ECS Fargate, RDS, CloudFront). Ran a security audit (unauthenticated payment endpoint, IDOR, sensitive data in logs, committed secrets) with a remediation plan; found the missing PayPal webhook that let subscriptions expire while customers kept being billed; then executed the client-approved production pause — full backups to S3, database kept as snapshots, service and load balancer shut down — and wrote the handover and relaunch guide.

Automated HR-to-Cloud Identity Provisioning Pipeline

ATH Infosystems

Azure Logic Apps pipeline syncing BambooHR to Microsoft Entra ID via the Microsoft Graph API and the SCIM bulkUpload endpoint, with OAuth2, Azure Key Vault, duplicate-conflict detection, automated group assignment and templated Office 365 welcome emails — a self-healing idempotent design that removed manual onboarding entirely.

Personal Projects

- **Portfolio website** ([Live](#) · [Code](#)) — my own site in HTML, CSS and JavaScript with Three.js / WebGL 3D scenes, case studies and a filterable project catalogue; responsive, accessible, hosted on GitHub Pages.
- **To-Do App with automated CI/CD** ([GitHub](#)) — deployed across 11 AWS EC2 instances via Ansible triggered by GitHub Actions. **Expense Tracker** ([Live Demo](#)) — Node.js, GraphQL, JWT, React. **Android chat app** ([GitHub](#)) — Kotlin, Firebase.